

Basic Mathematics

Lecture 1: Matrices

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Matrices $\longrightarrow$ Determinants $\longrightarrow$ Inversion $\longrightarrow$ Linear systems

Today: **read · combine · multiply · see structure**

- I 45 min Reading matrices and basic operations
- II 45 min Matrix multiplication
- III 45 min Special structure and row operations

Part I

Reading matrices

data

coefficients

relations

transformation

matrix = organised information

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \in \mathbb{R}^{m \times n}.$$

m rows

n columns

$$A \in \mathbb{R}^{m \times n}$$

shape first, calculation second

a_{ij}

first index: row

second index: column

$$M = \begin{pmatrix} 4 & -2 & 0 \\ 7 & 1 & 5 \end{pmatrix}$$

$$M \in \mathbb{R}^{2 \times 3}, \quad m_{11} = 4, \quad m_{12} = -2, \quad m_{23} = 5.$$

$$M = \begin{pmatrix} 4 & -2 & 0 \\ 7 & 1 & 5 \end{pmatrix}$$

first row

$$(4 \quad -2 \quad 0)$$

third column

$$\begin{pmatrix} 0 \\ 5 \end{pmatrix}$$

row

$$(a \quad b \quad c)$$

$$1 \times 3$$

column

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

$$3 \times 1$$

square

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$2 \times 2$$

rectangular

$$\begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}$$

$$2 \times 3$$

$$A = \begin{pmatrix} 2 & 1 \\ -1 & 3 \end{pmatrix}$$

data

coefficients

map $\mathbf{x} \mapsto A\mathbf{x}$

What is the shape?

Only then: which operation makes sense?

$A + B$ exists only if $\text{size}(A) = \text{size}(B)$.

$$(m \times n) + (m \times n) = m \times n$$

$$A = (a_{ij}), \quad B = (b_{ij})$$

$$A + B = (a_{ij} + b_{ij})$$

same position $\longleftrightarrow$ same position

$$\begin{pmatrix} 2 & -1 \\ 0 & 3 \end{pmatrix} + \begin{pmatrix} 5 & 4 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} 2+5 & -1+4 \\ 0+(-2) & 3+1 \end{pmatrix} = \begin{pmatrix} 7 & 3 \\ -2 & 4 \end{pmatrix}.$$

$$A - B = (a_{ij} - b_{ij})$$

$$\lambda A = (\lambda a_{ij})$$

both preserve the matrix shape

$$A = \begin{pmatrix} 2 & -1 \\ 0 & 3 \end{pmatrix}$$

$$-3A = \begin{pmatrix} -6 & 3 \\ 0 & -9 \end{pmatrix}.$$

$$\alpha A + \beta B$$

$$2 \begin{pmatrix} 1 & 0 \\ -1 & 2 \end{pmatrix} - \begin{pmatrix} 3 & 1 \\ 2 & 0 \end{pmatrix} = \begin{pmatrix} -1 & -1 \\ -4 & 4 \end{pmatrix}.$$

$$A + B = B + A$$

$$(A + B) + C = A + (B + C)$$

$$\lambda(A + B) = \lambda A + \lambda B$$

because these operations are entrywise

defined

$$(2 \times 3) + (2 \times 3)$$

$$5(4 \times 2)$$

not defined

$$(2 \times 3) + (3 \times 2)$$

$$(1 \times 4) - (4 \times 1)$$

- size of the matrix;
- meaning of a_{ij} ;
- rows versus columns;
- compatible entrywise operations.

Read the object first.

Part II

Matrix multiplication

Addition:

position + matching position

Multiplication:

row · column

$$(m \times n)(n \times p) = m \times p$$

inner dimensions match; outer dimensions remain

$$(2 \times 3)(3 \times 4) = 2 \times 4,$$

$$(1 \times 5)(5 \times 1) = 1 \times 1,$$

$$(4 \times 2)(2 \times 1) = 4 \times 1.$$

$(2 \times 3)(2 \times 4)$ is not defined

$$C = AB, \quad C = (c_{ij})$$

$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$$

row i of A meets column j of B

$$(a_{i1} \ a_{i2} \ \cdots \ a_{in}) \begin{pmatrix} b_{1j} \\ b_{2j} \\ \vdots \\ b_{nj} \end{pmatrix} = \sum_{k=1}^n a_{ik} b_{kj}$$

one row $\times$ one column = one number

$$A = \begin{pmatrix} 1 & 3 \\ -2 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 0 \\ 5 & -1 \end{pmatrix}$$

$$AB \in \mathbb{R}^{2 \times 2}$$

$$c_{11} = 1 \cdot 2 + 3 \cdot 5 = 17$$

$$c_{12} = 1 \cdot 0 + 3(-1) = -3$$

$$c_{21} = (-2) \cdot 2 + 4 \cdot 5 = 16$$

$$c_{22} = (-2) \cdot 0 + 4(-1) = -4$$

$$AB = \begin{pmatrix} 17 & -3 \\ 16 & -4 \end{pmatrix}$$

$$BA = \begin{pmatrix} 2 & 0 \\ 5 & -1 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ -2 & 4 \end{pmatrix} = \begin{pmatrix} 2 & 6 \\ 7 & 11 \end{pmatrix}.$$

$$AB \neq BA$$

$$\mathbf{x} \xrightarrow{B} B\mathbf{x} \xrightarrow{A} A(B\mathbf{x})$$

$$\mathbf{x} \xrightarrow{A} A\mathbf{x} \xrightarrow{B} B(A\mathbf{x})$$

different order = different process

$$A \in \mathbb{R}^{2 \times 3}, \quad B \in \mathbb{R}^{3 \times 1}$$

$$AB \in \mathbb{R}^{2 \times 1}$$

BA is not defined

$$\begin{pmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 0 & -1 \\ 4 & 2 \end{pmatrix} = \begin{pmatrix} 10 & 5 \\ 2 & -2 \end{pmatrix}.$$

$$(2 \times 3)(3 \times 2) = 2 \times 2$$

$$A \in \mathbb{R}^{m \times n}, \quad \mathbf{x} \in \mathbb{R}^{n \times 1}$$
$$A\mathbf{x} \in \mathbb{R}^{m \times 1}$$

input vector $\longrightarrow$ output vector

Let

$$A = (\mathbf{a}_1 \quad \mathbf{a}_2 \quad \cdots \quad \mathbf{a}_n), \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}.$$

Then

$$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n.$$

$$A\mathbf{e}_1 = \mathbf{a}_1, \quad A\mathbf{e}_2 = \mathbf{a}_2, \quad \dots, \quad A\mathbf{e}_n = \mathbf{a}_n.$$

columns show what the transformation does to basic directions

$$(AB)C = A(BC)$$

$$A(B + C) = AB + AC$$

$$(A + B)C = AC + BC$$

grouping may change; order may not

$$A^2 = AA, \quad A^3 = AAA, \quad A^0 = I$$

powers require a square matrix

- 1 read both shapes;
- 2 check inner dimensions;
- 3 predict the output shape;
- 4 compute row by column;
- 5 keep the order fixed.

$$(m \times n)(n \times p) = m \times p$$

Every output entry is one row–column interaction.

Part III

Visible structure and row operations

$$0_{m \times n} = \begin{pmatrix} 0 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 0 \end{pmatrix}$$

$$A + 0 = A$$

$$I_n = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$$

For $A \in \mathbb{R}^{m \times n}$:

$$I_m A = A, \quad A I_n = A.$$

left identity: $m \times m$ right identity: $n \times n$

$$A = (a_{ij}) \in \mathbb{R}^{m \times n}$$

$$A^T = (a_{ji}) \in \mathbb{R}^{n \times m}$$

rows become columns

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}^T = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}.$$

$$2 \times 3 \longrightarrow 3 \times 2$$

$$(A^T)^T = A$$

$$(A + B)^T = A^T + B^T$$

$$(AB)^T = B^T A^T$$

transpose reverses multiplication order

$$A^T = A$$

$$\begin{pmatrix} 2 & -1 & 4 \\ -1 & 3 & 0 \\ 4 & 0 & 5 \end{pmatrix}$$

mirror symmetry across the main diagonal

$$D = \text{diag}(d_1, d_2, \dots, d_n)$$

$$D = \begin{pmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{pmatrix}.$$

$$D = \text{diag}(3, -1, 2)$$

$$D^2 = \text{diag}(9, 1, 4)$$

$$D^3 = \text{diag}(27, -1, 8)$$

no mixing between coordinates

$$D_1 = \text{diag}(a_1, \dots, a_n), \quad D_2 = \text{diag}(b_1, \dots, b_n)$$
$$D_1 D_2 = D_2 D_1 = \text{diag}(a_1 b_1, \dots, a_n b_n).$$

Look for:

- many zeros;
- equal rows or columns;
- proportional rows or columns;
- diagonal or triangular patterns;
- symmetry;
- repeated blocks.

- ① swap two rows;
- ② multiply a row by a non-zero number;
- ③ add a multiple of one row to another.

controlled transformations, not random changes

$$R_i \leftrightarrow R_j$$

$$R_i \leftarrow \lambda R_i, \quad \lambda \neq 0$$

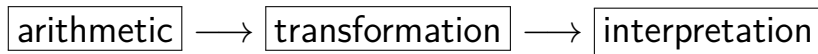
$$R_i \leftarrow R_i + \lambda R_j$$

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 2 & 4 & 3 \\ 0 & 1 & 5 \end{pmatrix}$$

$$R_2 \leftarrow R_2 - 2R_1$$

$$(2, 4, 3) - 2(1, 2, -1) = (0, 0, 5)$$

$$\begin{pmatrix} 1 & 2 & -1 \\ 2 & 4 & 3 \\ 0 & 1 & 5 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 2 & -1 \\ 0 & 0 & 5 \\ 0 & 1 & 5 \end{pmatrix}.$$



the new row reveals hidden structure

$$\mathbf{u} = \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix} \in \mathbb{R}^{3 \times 1}$$

$$\mathbf{u}^T = (2 \quad -1 \quad 4) \in \mathbb{R}^{1 \times 3}$$

$$(1 \times 3)(3 \times 1)$$

$$\mathbf{u}^T \mathbf{u} \in \mathbb{R}^{1 \times 1}$$

$$(3 \times 1)(1 \times 3)$$

$$\mathbf{u} \mathbf{u}^T \in \mathbb{R}^{3 \times 3}$$

For $\mathbf{u} = (2, -1, 4)^T$:

$$\mathbf{u}^T \mathbf{u} = 2^2 + (-1)^2 + 4^2 = 21$$

$$\mathbf{u} \mathbf{u}^T = \begin{pmatrix} 4 & -2 & 8 \\ -2 & 1 & -4 \\ 8 & -4 & 16 \end{pmatrix}.$$

- ① identify every matrix shape;
- ② decide which operations exist;
- ③ predict the result shape;
- ④ use the correct local rule;
- ⑤ exploit visible structure;
- ⑥ interpret the result.

avoid

- mixing row and column indices;
- adding different shapes;
- multiplying entrywise;
- swapping multiplication order.

replace with

- read a_{ij} carefully;
- check dimensions first;
- use row–column products;
- preserve the order.

- What is the shape?
- What does each index mean?
- Which operations are defined?
- What is the output shape?
- Does order matter?
- What structure can I see?

Read the shape.

Then calculate.